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Efficient conversion of benzene and syngas to toluene and xylene over ZnO-ZrO₂&H-ZSM-5 bifunctional catalystsXiao Zhao¹, Xuan Shi¹, Zhongshun Chen¹, Long Xu^{1,2}, Chengyi Dai^{1,2,*}, Yazhou Zhang¹, Xinwen Guo³, Dongyuan Yang^{4,5,*}, Xiaoxun Ma^{1,2,*}¹ School of Chemical Engineering, Northwest University, Xi'an 710069, China² International Science & Technology Cooperation Base for Clean Utilization of Hydrocarbon Resources, Chemical Engineering Research Center of the Ministry of Education for Advanced Use Technology of Shanbei Energy, Collaborative Innovation Center for Development of Energy and Chemical Industry in Northern Shaanxi, Northwest University, Xi'an 710069, China³ School of Chemical Engineering, Dalian University of Technology, Dalian 116024, China⁴ Shaanxi Key Laboratory of Energy Chemical Process Intensification, School of Chemical Engineering and Technology, Xi'an Jiaotong University, Xi'an 710049, China⁵ Shanxi yanchang, Petroleum (Group) Corp. Ltd, Xi'an 710000, China

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ABSTRACT

A series of ZnO-ZrO₂ solid solutions with different Zn contents were synthesized by the urea co-precipitation method, which were coupled with H-ZSM-5 zeolite to form bifunctional catalysts. As a new benzene alkylation reagent, syngas was used instead of methanol to realize the efficient conversion of syngas and benzene into toluene and xylene. A suitable ratio of ZnO-ZrO₂ led to the significant improvement in the catalytic performance, and a suitable amount of acid helped to increase the selectivity of toluene/xylene and reduce the selectivity of the by-products ethylbenzene and C₉ aromatics. The highest benzene conversion of 89.2% and toluene/xylene selectivity of 88.7% were achieved over 10% ZnO-ZrO₂&H-ZSM-5 (Si/Al = 23) at a pressure of 3 MPa and a temperature of 450 °C. In addition, the effect of the zeolite framework structure on product distribution was examined. Similar to the molecular dynamics of aromatic hydrocarbons, H-ZSM-5 zeolites comprise 10-membered-ring pores, which are beneficial to the activation of benzene; hence, the conversion of benzene is higher. H-ZSM-35 and H-MOR zeolites exhibited small eight-membered-ring channels, which were not conducive to the passage of benzene; hence, the by-product ethylbenzene exhibits a higher selectivity. The distance between the active centers of the bifunctional catalysts was the main factor affecting the catalytic performance, and the powder mixing method was more conducive to the conversion of syngas and benzene.

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1. Introduction

Aromatics are the most basic raw materials of the organic chemical industry, which are widely used in the synthesis of resins, fibers, rubber, dyes, and other chemical products [1–3]. Aromatics mainly originate from petroleum and coal tar, and light aromatics such as benzene, toluene, and xylene (BTX) are the most important products, accounting for an important share in the petrochemical industry. Currently, about 70% of the global supply of BTX originates from naphtha cracking and catalytic reforming [4–7]. How-

ever, with the increasing shortage of petroleum resources, the technology of synthesizing light aromatics from non-petroleum carbon resources has gradually become an important problem that needs to be solved. Currently, syngas has become an important resource to replace petroleum synthetic chemical products. Syngas can be obtained from widespread sources such as natural (shale) gas, coal, biomass, and carbon-containing waste, as well as carbon dioxide conversion [8–11]. It is one of the important platforms for the use of non-oil resources to supply energy and chemical raw materials.

Conventionally, toluene/xylene is produced from benzene and syngas first via the synthesis of methanol by the selective hydrogenation of CO over a metal catalyst (such as CuZnAlO_x [12], ZnZrO_x [13], ZnAlO_x [14], etc.), followed by the alkylation of methanol and benzene on a zeolite catalyst [15–17]. Although the above two-step technical route is relatively mature, the following issues

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